1545 — ① ← Medium

Find K^{th} Bit in N^{th} Binary String

$S_1 = '0'$

$S_i = S_{i-1} + "1" + reverse(invert(S_{i-1}))$ $\quad i > 1$

$S_2 \to 0 \to 1$

$S_2 = S_1 + "1" + rev(invert(S_1))$

$= S_1 + "1" + rev("1")$

$= "0" + "1" + "1"$

$\Rightarrow "011"$

$0 \to 1$
$1 \to 0$

1234
4321

$S_3 = S_{3-1} + "1" + rev(invert(S_{3-1}))$

$= S_2 + "1" + rev(invert(S_2))$

$S_2 = 011$
$\downarrow\downarrow\downarrow$
100

$\Rightarrow S_2 + "1" + rev("100")$

$\Rightarrow "011" + "1" + "001"$

$\Rightarrow "0111001"$

n, K → 2

3

K = 4

$S_4 = S_3 + "1" + rev(invert(S_3))$

0111001
1000110
↓
0110001

0111001 + 1 + 0110001

1 0111001011001

K = 5

① K = 1 idx
6 x

⇒ First Approach :-

Hm String banate hai , (n)

~~First~~ Return $(k-1)^{th}$ char

= 0 ✗

1 ✓

" 0111001 "

1

String banegi Kese ?

$n=3$

$S_1 =$ "0"

newStr = "0" + "1" + prev (invert (str prev))

Invert :- prev string 0 → '1'

1 → '0'

Reverse :- prev = "0111001"

↓

'1001110'

⇒ '0111001' → 1000110 → 0110001

$n = 3$

0

Algo/Flow :-

- Create StringBuilder sb, n--
- while n > 0
  - Dummy StringBuilder sb2 = new prev
  - Append "1" in dummy
  - Apply formula
  - Travel sb Right to left
  - if 0 → 1 append sb2
    1 → 0 append sb2
  - sb = sb2
  - n--

prev = "0"

n = 3

dum = 0 + 1 + 2 + 3 =

n

- sb (K-1) chan

$1 + 3 + 7 + 15 + \dots 2^n$

T.C ⇒ 2^n → O(n)

S.C ⇒ 2^n → O(n)

Previor Approch Mistake ?

- while pura Strg banana
  - usko rev
  - invert

- Pehla Sawal ?

		len
n = 1	"0"	1
n = 2	"011"	3
n = 3	"0111001"	7
n = 4	"01110011011001"	15

2^2

$2^4 = 16-1$

$2^n - 1$

•> Optimization

Step 1 Pori string banana need X $\frac{100}{\text{"1"}}$ (k^n)

↓

Kya k^{th} position ko mathematically derive kr sake hu

$$\boxed{S_n = (S_{n-1}) + (\text{"1"}) + (\text{rev}(\text{invert}(S_{n-1})))}$$

mid =

n=2 → $S_1 \Rightarrow \underset{S_1}{0}$ 1 $\underset{S_1}{1}$ | n=3 → S_2 1 S_2

n=4 S_3 1 S_3 | n=7 S_6 1 S_6

•> length $2^n - 1$ → $(1 << n) - 1$

mid = $\left(\frac{len}{2}\right) + 1$

n=3, 1 2 3 | n=4 1 2 3 4 | n=5 1 2 3 4 5 (F)

mid = $\frac{3}{2} + 1 = 2$ | mid = $\frac{4}{2} + 1 = 3$ | $\frac{5}{2} + 1$ 3

Step 2 Ab sabse important Observation :-

$$\left[S_n = (S_{n-1}) + (\text{"1"}) + (\text{rev}(\text{invert}(S_{n-1})))\right]$$

Pos (K=)

•> 1 ... mid-1

mid

mid+1 ... end

Belong To

left part (S_{n-1})

1

Right Part (rev + invert of S_n)

n=7

S_6 1 (1) ... 3 & 8

• Finally Agar K == middle → 1

K < middle → left

K > middle → right

•> K < middle

"011" 011 001 7/2 9

011 1 001 7 1,2,3 K 1,2,3 5,6,7

•> K > middle: Right part ko ?

Right side idx —relate→ Left side idx

totalLen = $2^n - 1$

mid = totalLen / 2

Agar K > mid

Right part → K - mid

ex1 len = 7

011 001 1 2 3 4 5 6 7

n=3 K=5, 6 mid 7 5

"011"

5 ↔ 3

6 ↔ 2

7 ↔ 1

total L = 7

K > mid

newK ⇒ L-K+1

7 - 5 + 1 = 3

7 - 6 + 1 ⇒ 2

7 - 7 + 1 = 1

$n = 3$

1 2 3 4 5 6 7

5 → 3
6 → 2
7 → 1

k = 5	k = 6	k = 7
L = 7	7 − 6 + 1	7 − 7 + 1
7 − 5 + 1 = 3	⇒ 2	1

⇒ newK = L − K + 1 → Reverse map

"011" 1 "001"

5

3

n = 2,

⊙ Find K > mid :-

n = 2

K = 1 2 3

newK = L − K + 1

n = 2,

Ans = invert (findKBit (n − 1, newK))

"001"

n = 2, K = 3

0 1

(1 << 1) − 1

⇒ Base Case :- n == 1 , 0 → 3

L − K + 1

3 − 3 + 1

S1 → "0" 2

(1, 1)

n = 1

n = 2

K = 1, 2, 3

⇒ Flow n == 1 , "0"

0 1

if K == mid → "1"

K < mid → solve (n − 1, K)

K > mid → solve (n − 1, L − K + 1) × invert

1

1

2

K = 2

Algo 1:

1. Base Case
   if $n == 1 \rightarrow$ '0'
2. Find Middle Position
   TotalLen $= 2^n - 1$
   middle Idx $= (\text{Totallen}/2) + 1$
3. if $k == \text{mid} \longrightarrow$ '1'
4. if $k < \text{mid}$
   Ans lie left part
   Rec $\rightarrow$ ans$(n-1, k)$
5. $k > \text{mid}$
   - Ans lie right part
   - Find mirror idx
     newk $=$ totallen $- k + 1$
   - RevCall
     Ch $=$ ans$(n-1, \text{newk})$
   - Invert $0 \rightarrow 1$
     $1 \rightarrow 0$

$n = 3 \quad 2 \quad 1$

$n-3, 2$

$O(n)$

T.C $= O(n)$

S.C $= O(n)$

$O(1)$